

Probability and Statistics

SITAMS CSE (AI & ML) • R23 • II Year • IV Semester • Course Code: 23BSC243T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: DESCRIPTIVE STATISTICS

Syllabus topics

- Statistics Introduction, Population vs Sample, Collection of data, primary and secondary data, Measures of Central tendency, Measures of Variability (spread or variance) Skewness, Kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Source page(s): 114

Unit 2: PROBABILITY

Syllabus topics

- Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Bayes's theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation

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Unit 3: PROBABILITY DISTRIBUTIONS

Syllabus topics

- Probability distributions: Binomial, Poisson and Normal-their properties (Chebyshevs inequality)
- Approximation of the binomial distribution to normal distribution

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Unit 4: ESTIMATION AND TESTING OF HYPOTHESIS, LARGE SAMPLE TESTS

Syllabus topics

- Estimation-parameters, statistics, sampling distribution, point estimation, Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors and power of the test
- Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means
- Confidence interval for parameters in one sample and two sample problems

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Unit 5: SMALL SAMPLE TESTS

Syllabus topics

- Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), χ^2 - test for goodness of fit, χ^2 - test for independence of attributes

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Study Hub status

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